

Lab 5

(1). dependencies I used and why.

SQLite: I used it because it is the plugin that lets flutter create and interact with SQLite database.

path: I used it because the correct file path for saving the database on the device.

The reason for SQLite is because it is necessary for the app; it needs to work offline. It works to store data directly on the device, supports data with multiple fields, and allows for CRUD operations which shared preferences cannot do.

(2). how I handled database exceptions:

* Database not initialized: I call initDB() before any operation ~~in a try-catch~~ ~~where the database~~ and wrap it in a try-catch. if it fails I show the user an error message.

* insert/update/delete errors: I wrap every CRUD operation in a try-catch block to catch DatabaseException and display a friendly error instead of crashing.

* Invalid or corrupted data: I validate inputs before saving checking that fields are not empty. I also use null checks when reading data and if the database is corrupted I catch the error on openDatabase() and offer a reset option.

(3) SQLite in flutter:

Database Vs Table: the database is the file stored on the device (eg: posts.db). the table lives inside ~~the~~ it and holds the actual rows of data.

My app has one table called (posts) with column id, title, and body.

• CRUD operations

- Create $\rightarrow$ `db.insert('posts', post.toMap())`
- Read $\rightarrow$ `db.query('posts')`
- update $\rightarrow$ `db.update('posts', post.Map(), where: 'id=?')`
- Delete $\rightarrow$ `db.delete('posts', where: 'id=?')`

Asynchronous Interaction: Database operations take time so flutter uses async/await with future to avoid blocking the UI. After the operation completes I call setState to refresh the screen.

Offline Posts Manager

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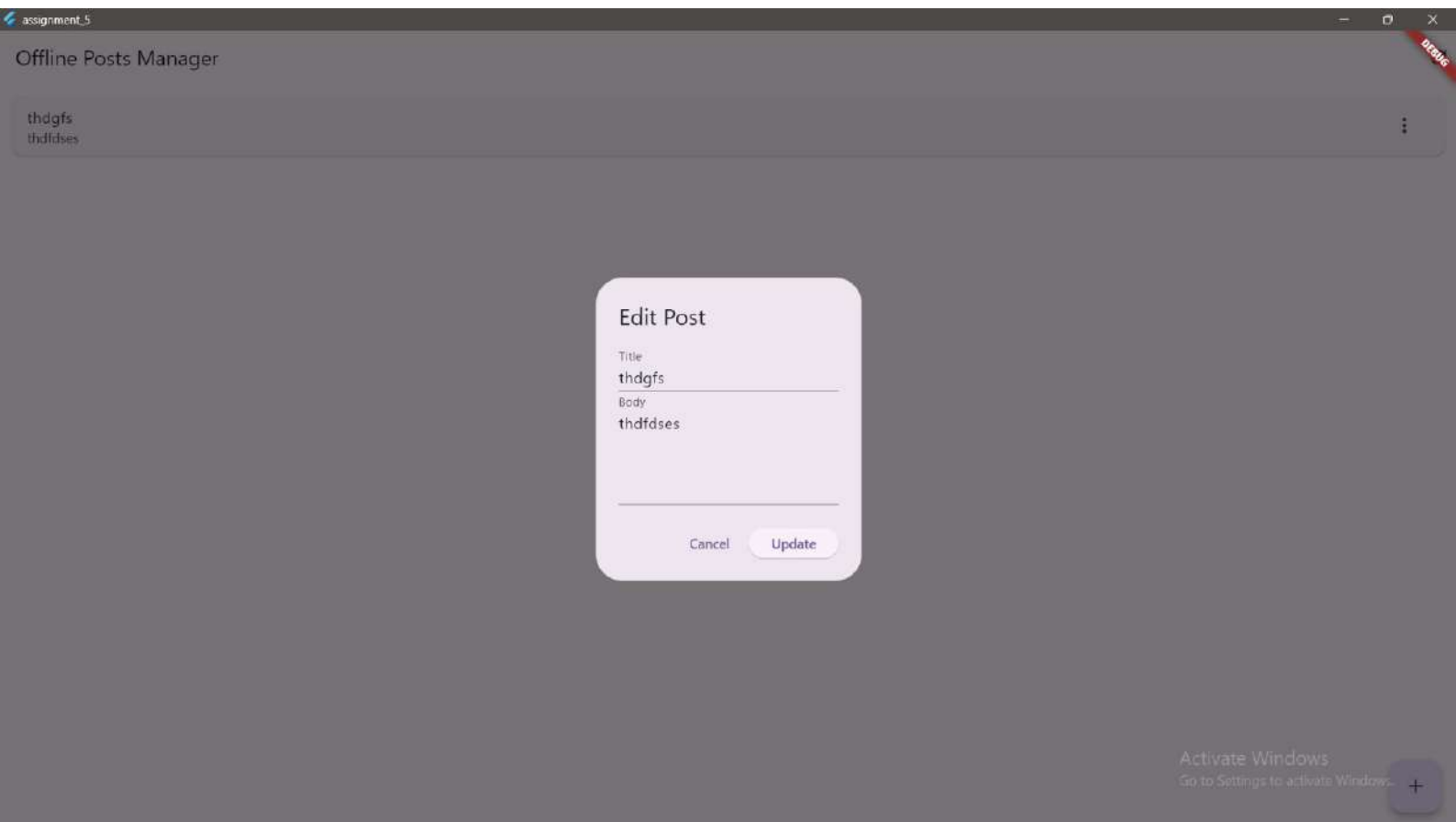
Offline Posts Manager

thdgfs thdfdses	View Edit Delete
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Title

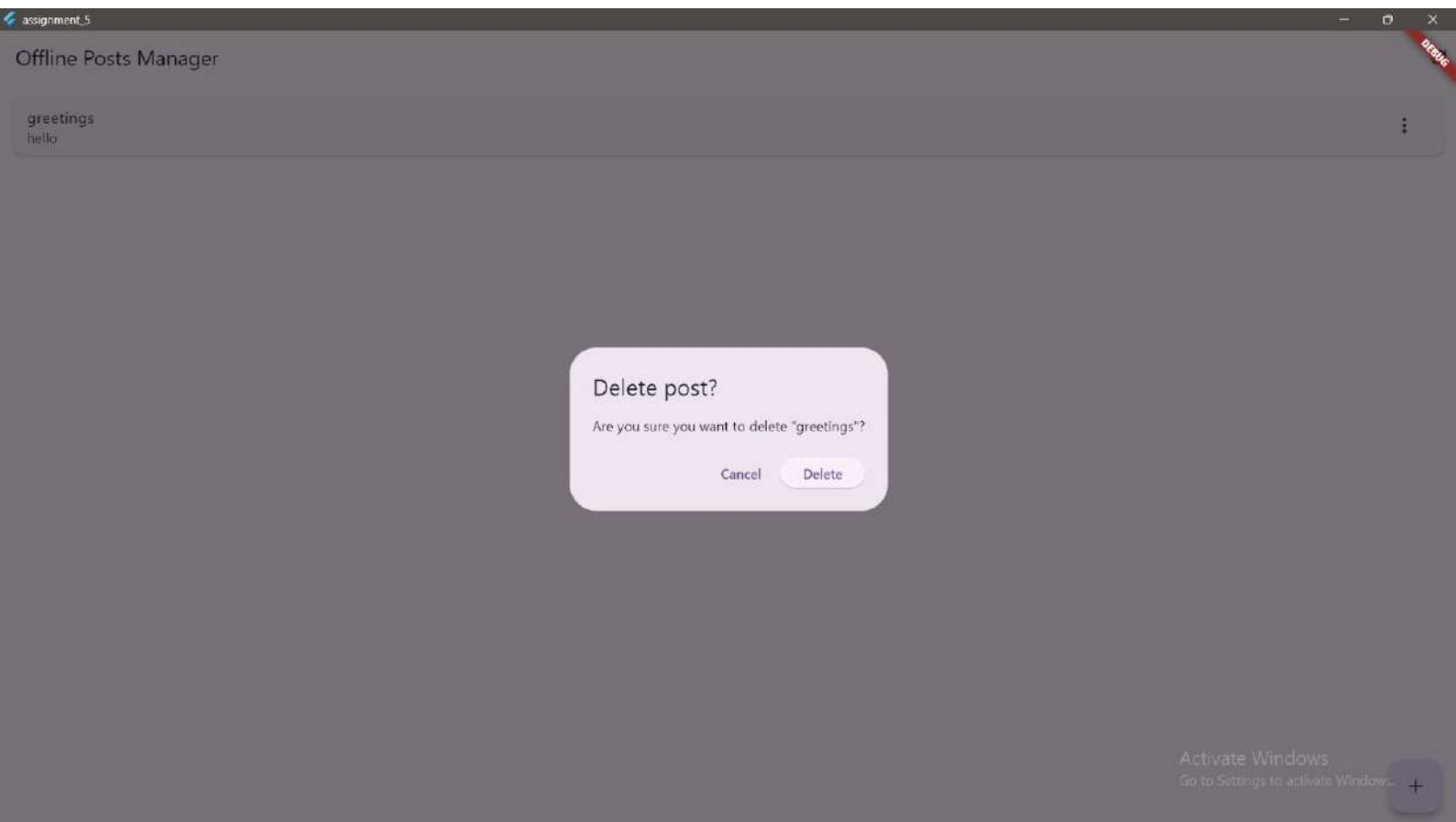
Body
thdfdses

Activate Windows
Go to Settings to activate Windows.



Offline Posts Manager

greetings
hello



Offline Posts Manager

greetings
hello

Delete post?

Are you sure you want to delete "greetings"?

Cancel Delete

Offline Posts Manager

No posts yet. Add one using + button.

Activate Windows
Go to Settings to activate Windows.

